AAE 554 - Fatigue of Structures and Materials
3 credits

1 Course Information/Description

Development and application of methods for predicting the fatigue life of structural components. Characterization and response of materials to cyclic loading. Fatigue resistant design of aerospace structures. Both fatigue crack initiation and crack propagation concepts are discussed.

2 Learning Resources and Reference Text

Brightspace Learning Management System:
Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.
Textbook:

- **Highly Recommended Textbook**
  - *Metal Fatigue in Engineering*, Ali Fatemi, Wiley, 9780471510598 *(available online via Purdue Libraries in Brightspace)*
- **Other Supplementary Readings**
  - *Mechanical Behavior of Materials,* Norman E. Dowling, Prentice Hall *(a print copy may be available in the Engineering library)*
  - *Fatigue of Materials,* Subra Suresh, Cambridge *(available online via Purdue Libraries and in Brightspace)*
  - *Fundamentals of Structural Integrity,* A.F. Grandt, Jr., Wiley *(a print copy may be available in the Engineering library)*
- Additional relevant journal papers will be posted to Brightspace

3 Assignments

Your learning will be assessed through a combination of weekly homework and two exams spread throughout the academic period. Details on these assignments and exams, including a schedule of due dates will be posted on the course website.

- Homework (10 total, 25%, ongoing) - Assigned weekly on the previous Tuesday (given on Brightspace) and due on the following Thursday by 10:00 AM ET via Brightspace. Late HW will not be accepted.
  - The HW is for your own benefit and is necessary to properly learn the material. It is expected that each student put forward an honest effort in solving each problem. Although discussion of assignments between students is permitted, the solution should represent your own efforts and understanding of the problem. Blatant copies of the solutions will not be accepted, and repeated evidence of copying will lead to failure of the course.
    - In most cases, homework will be graded within 1-2 weeks after the due date of the submission. In most cases, the answer key will be available within a week of the submission.

- Lastly, the purpose of homework assignments is for students to gain further understanding of classroom principles through application to practical problems. A critical aspect is problem solving and thinking about strategies to solve each question. This training is important to your overall understanding of the material and the ability to do well on the exams.
- HW will be graded based on approach, demonstration of work, and effort (not necessarily on the answer). Full credit will be given if the student has demonstrated that thought and effort has been put into the solution and the student is on the right track. If the question is not attempted, minimal effort is given, the student gives a flippant response, or the trajectory towards the solution is completely off path, then no credit will be given for this question. The individual HW will not be marked up to identify where the student deviated from the path and give partial credit, as we would rather dedicate this time to spending time with students a priori and towards teaching. Instead, a detailed HW solution will be given to the students after the HW is submitted. This HW solution can be used as a study guide towards the exam.
- 11 HWs will be assigned, only the highest 10 scores will count towards your grade. In other words, students are allowed to drop 1 HW score. This is intended, so students can manage their other responsibilities for work, other class projects, research, travel, etc. that might come up during the semester. As well, dropping the lowest HW may ease some tension about mistakes with HW submissions (forgot, uploaded the wrong assignment, missed a deadline, etc.).

- Midterms (1 total, 35%, Tuesday 10/21, 8-10 pm, MSEE B012) - The midterm will be held at night for on campus students, to allow students more time and space. For EPE students, they must arrange for a proctor in accordance to EPE rules. EPE students must take the exam +/- 1 day of the on-campus students.
  - A non-communicating calculator and 8.5”x11” crib sheet are allowed. The crib sheet must be hand-written and cannot be duplicated and used by another student. The material on the exam will be closely related to the lectures and types of questions asked on the HW.
  - The exam solutions will not be distributed to the students but will be reviewed in class. Exam grades will not be openly discussed. If the students have grade appeals, they must submit a written appeal along with the original copy of their graded exams within one week of the exams being returned. The exam will be re-graded, which may reduce the overall grade, as any grading that was too generous may be corrected.
- Final Exam (1 total, 40%, DATE TBD) – The exam will be accumulative. The final exam date/time is determined by the registrar. For distance students, they must arrange for a proctor in accordance to EPE rules. Distance students must take the exam +/- 1 day of the on-campus students.
  - A non-communicating calculator and 8.5”x11” crib sheet are allowed. The crib sheet must be hand-written and cannot be duplicated and used by another student. The material on the exam will be closely related to the lectures and types of questions asked on the HW.

4 Course Learning Objectives

By the end of the course, you will be able to:

- Develop methods for characterizing fatigue resistance of materials

- Predict cyclic lives of structural components
- Apply approaches for preventing fatigue failures through materials selection, fatigue resistant design, and "fleet management" concepts
- Examine crack initiation and crack propagation approaches for life assessment

Video Title	**Learning Objectives**	**HW**
1 - Introduction		
Intro into the concept of fatigue	Recall definition of fatigue	
Historical Origins of fatigue	Recall historical origins of fatigue through case studies	
Cost of fatigue	Demonstrate the economic impact of fatigue	
Fatigue in Aerospace Components	Illustrate the implications and severity of fatigue failures in aerospace components	
Applications of Fatigue to Composites	Assess the challenges of fatigue analysis in composites	
2 - Design for Fatigue		
Overview of Component Life Assessment	Interpret the flowchart for fatigue life decisions of aerospace components	
Design Requirements and Structural Analysis	Infer geometric, load, environment, material, and processing constraints Relate stress analysis of components to stress hotspot locations	
HCF vs LCF	Contrast high cycle fatigue and low cycle fatigue	
Design Life Models - Safe Life	Classify stress and strain based approaches to calculate safe life	
Design Life Models - Damage Tolerance	Recall existance of pre-existing damage Evaluate fatigue crack growth characteristics	
Review Design Life Models	Compare/Contrast (i) high cycle fatigue, (ii) low cycle fatigue, and (iii) damage tolerant approaches	
Spectrum Loading	Utilize accumulation of damage to apply to spectrum loading	HW 1
Manufacturing Influence on Fatigue	Identify manufacturing characteristics that influence fatigue behavior	
In Service Useage	Identify in service degredation and variables leading to repair, life extension, retirement	
Other Considerations for Fatigue	Relate more complex fatigue phenemena, which will be covered in future lectures	
3 - High Cycle Fatigue		
Overview of High Cycle Fatigue	Recall key aspects of safe life approaches, including infinite life Distinguish key characteristics of high cycle fatigue versus low cycle fatigue	
Mechanics of rotating bending tests	Interpret mechanics of rotating bending testing	
Unixaxial HCF testing	Interpret mechanics of unixial testing, including specimen design from ASTM standards	
Stress-Life Plots	Construct a stress-life curve Identify the endurance limit Interpret variability (scatter) in life	
Basquin's Law	Identify the power law fit (Basquin's law) to the stress-life curve Analyze Basquin parameters from curve or table	
Mean Stress	Identify mean stress effect on stress-life curve Define stress amplitude, mean stress, stress ratio.	

Goodman Diagram	Construct Goodman diagram and apply Goodman's equation	
Example problem - Goodman Mean Stress Correction	Calculate life, based on mean stress correction using Goodman equation	
Overview of Mean Stress Correction Models	Recall and apply rationale for mean stress correction for high cycle fatigue	
Various Mean Stress Correction Models	Evaluate Soderberg, Morrow, Goodman, and Gerber mean stress models	HW 2
Origin of Notch Effects	Identify physical and geometric origins of stress concentrations leading to notch effects	
Stress Concentrations	Determine stress concentrations from analyical and chart approaches	
Fatigue Notch Sensitivity	Justify rationale and physical origins of fatigue notch factor Calculate the notch sensitivity parameter	
Characteristic length scale for notches	Apply characstic length scale (notch root radius) related to strain gradients to notch sensitivity	
Surface roughness influence on HCF	Recall surface roughness for common manufaturing methods as well as means to measure surface roughness Establish approximate fatigue knock down factor based on surface roughness	
Metrics of surface roughness	Determine common metrics for surface roughness (1st, 2nd, 3rd, 4th order moments) and relate to fatigue performance	
Tribology and fretting	Define fretting failure and tribology effects of surfaces	
Other factors influencing HCF	Interpret environmental, microstructure, frequency, and size effects on high cycle fatigue	
Overview of Residual Stress Effects	Evaluate residual stress effect on high cycle fatigue	
Origins of Residual Stresses	Interpret mechanical, thermal, manufacturing, plating as it relates to residual stress effects	
Measuring Residual Stresses	Summarize destructive, non-destructive, and analytical means to measure residual stress	
Analysis using Residual Stresses	Interpret and analysis residual stress effects	
Shot Peening	Infer shot peening as a controlled cold work process to impart residual stress and improve fatigue life	
Shot Peening Q&A	Relate shot peening to aerospace components	HW 3
Variability in HCF Data and Normal Distributions	Classify variability/scatter based on life and/or stress for high cycle fatigue Evaluate key characteristics of a normal distribution\	
Coefficient of Variance	Determine coefficient of variance (COV) and categorize COV based on mechanical property	
Fatigue Probability of Failure Plots	From stress-life data, construct and interpret a probability of failure plot	
Identifying the -3sigma minimum	Calculate the -3 sigma minimum life for high cycle fatigue	
Material rationale for variability	Evaluate the materials rationale for variability in high cycle fatigue life	
Reliability Assessments	Compare deterministic versus probablistic approaches Recall sources of variability and uncertainty Analyze reliability based on many factors influencing life	
DARWIN	Recall DARWIN as a tool for probablistic damage assessment based on zoning	
4 - Low Cycle Fatigue		
Controlling Strain during LCF	Examine need and methods for strain controlled tests	
Overview of Bauschinger Effect	Relate the Baushinger effect and importance to fatigue	
Yield Surface Defintion	Construct a yield surface	

Types of Hardening	Classify types of hardening (kinematic and isotropic)	
Isotropic Hardening	Evaluate evolution of yield surface based on isotropic hardening	
Kinematic Hardening - Bauschinger Effect	Evaluate evolution of yield surface based on kinematic hardening Deduce origins of Bauschinger effect based on kinematic hardening	
Cyclic Hardening/Softening	From hysteresis loops, evaluate cyclic softening, hardening, or a combination	
Hyersteresis Loop Evaluation	Calculate elastic and plastic strain range from hysteresis loops	
Ramberg-Osgood Model for Montonic Loading	Construct the Ramberg-Osgood fit of a uniaxial tension	
Cyclic Stress Strain Curve	Construct a cyclic stress-strain curve Identify test methods to create a cyclic stres-strain curve	
Ramberg-Osgood Model for Cyclic Loading	Determine a Ramberg-Osgood fit of cyclic stress strain-curve	HW 4
Strain-Life Approach	Establish strain - life plots for low cycle fatigue	
Coffin-Manson Eq and Unified Strain Approach	Determine the elastic and plastic strain components versus life model Calculate regression fits of the power law model	
Shape of the Hystersis Loop	Relate the shape of the stable hysteresis loop, when elastic and plastic strain curves intersect	
Mean Stress Effect on LCF	Relate to models adjusting strain-life approach for a given mean stress Identify rationale for mean stress having a greater influence on high cycle fatigue regime	
Mean Stress Relaxation	Explore mean stress relaxation as a form of kinematic hardening	
Shakedown and Ratcheting	Distinguish hysteresis loop evolution and plasticity rationale for shakedown and ratcheting	
Masing Behavior and Plastic Strain Energy Definition	Differentiate Masing vs Non-Masing behavior For Masing behavior, calculate the plastic strain energy associated with the integration of the hysteresis loop	
Morrow's Plastic Strain Energy Analysis	Recall Morrow's study of plastic strain energy, including trends and limitations	HW 5
Notch Effects in LCF	Identify notch effects related to low cycle fatigue Identify stress and strain concentration factors near a notch	
Neuber Approach to Notches	Calculate Neuber's stress concentration and associated life related to notches	
Glinka's Approach to Notches	Calculate Neuber's strain energy density and associated life related to notches	
Plane Stress / Plane Strain Application to Notches	For size of notch, relate plane stress vs plane strain denotation and application of multixaxial loading	
5 - Physics of Fatigue		
Length scales associated with fatigue	Illustrate why fatigue is a multi-length scale issue	
Crack evolution regimes of the fatigue process	Classify stages of the fatigue process	
Material/Crystal Structure and Slip System Introductions	Classify types of material structures and common crystal (BCC, FCC, HCP) structures Recall Miller indices denote planes and directions Relate crystallograhic slip to specific sets of crystallographic planes/directions	
Defects in Crystals - Point Defects	Classify types of point defects in crystals	
Introduction to Dislocations	Extend dislocation motion to rationalize plastic deformation and hardening behavior Classify types of dislocations	

Forces on Dislocations	Interpret forces on dislocations Identify minimum energy configurations of dislocations based on force interactions	
Motion of Dislocations	Extend and explain the dislocation motion in metails related to the mechanism of plasticity	
Planar Defects - Grain and Twin Boundaries	Identify the role of grain/twin boundaries in defining a polycrystalline microstructure	
Volume Defects - Precipitates	Classify the type of precipitates	
Strengthening Mechanisms in Metals	Construct the type of strengthening mechanisms in metals and their contribution to strength	
Crystallographic Slip and Schmid Factor in Single Crystals	Calculate the resolved shear stress Interpret crystallographic slip based on orientation	
Deformation of polycrystals	Examine characteristics of deformation in polycrystalline alloys Relate deformation in polycrystalls to strain localization	
Fatigue Preliminaries	Identify key characteristics of cyclic deformation Relate sequence of fatigue mechanisms to crack initiation	
Slip Bands	Identify slip band formation in fatigue of polycrystalline materials	
Intrusions and Extrusions	Relate concepts of slip to the formation of intrusions/extrusions	
Persistent Slip Bands (PSBs)	Define persistent slip bands (PSBs)	
Low Energy Configurations in PSBs	Relate unique (low energy configuration) dislocation structures to PSBs	
Models of PSBs	Interpret classical PSB fatigue models (based on surface roughness, hardening, PSB-GB interaction) to crack initiation	
Sources of Irreversibilities	Interpret types and mechanisms of irreversibilities to cyclic loading	HW 6
Crack initiation in engineering alloys	Apply microstructure rationale to sources of crack initiation in engineering alloys	
6 - Multiaxial Fatigue		
Motivating Multiaxial Stress States Studies	Identify complex stress states resulting from geometric features on components	
States of Stress	Explain stress tensor and 3D state of stress from common loading	
Equivalent Stress Approach	Calculate equivalent stress and apply to stress-based fatigue approaches	
Stress-based Approach	Analyze Sines and Findley based stress approaches	
Non-proportional Loading for Stress-based Approaches	Interpret non-portional loading Interpret stress-based approaches application to non-proportional loading	
Critical Plane Approaches	Explain physical basis for critical plane approaches Apply critical plane approaches for life determination	
Multiaxial Stresses Applied to Notches	Interpret multiaxial stress state for notches	
7 - Fatigue Crack Growth		
Overview of damage tolerant approaches	Demonstrate need for damage tolerant approaches, based on acknowledgement of pre-existing damage	
Historical rationale via past failures	Recall past historical case studies neccessitating damage tolerant approaches	
Damage Tolerance Rationale	Contrast damage tolerant approaches to safe life approaches Recall regimes of fatigue (crack) processes	
Griffith approaches to fracture	Relate released elastic strain energy to crack growth	

Stress fields around a crack and stress intensity	Identify stress fields around a crack Classify modes of loading related to a crack Construct the stress intensity factor Define concept of fracture toughness	
Crack tip plasticity	Explain rationale for plasticity of the crack tip Distinguish plastic zone ahead of a crack tip	
Size effects for fracture toughness	Identify geometric constraints on plastic zone formation Relate specimen/crack size to plastic zone size	HW 7
Fracture toughness - macro and micro perspective	Explain atomistic mechanism of plastic formation Relate plastic zone formation to toughening mechanism	
Effects of temperature on fracture toughness	Relate temperature dependency on fracture toughness	
Fracture resistant design	Determine maximum safe load of a given defect Determine maximum allowable defect size of a given load	
Linear Elastic Fracture Mechanics	Apply linear elastic fracture mechanics to evaluate strength of a cracked structure	
Geometric correction factor	Identify the need for a geometric correction factor on the stress intensity factor Determine the geometric correction factor of simple geometries from charts or analytical expressions	
Damage Tolerance Adoption and Design Requirements	Summarize the USAF adoption of a damage tolerant approach to lifing aircrafts	
Fatigue crack growth characteristics	Identify beachmarks and striations as evidence of fatigue crack propagation Distinguish safety by inspection (repair) protocols	
Tests to capture fatigue crack growth rates	Determine testing protocols and analysis to measure the fatigue crack growth rates	
Regions of Fatigue Crack Growth	Differentiate the characteristics of the three regions of fatigue crack growth	
Threshold stress intensity factor	Calculate the threshold stress intensity factor Design for non-propagating cracks	
Steady State Crack Growth and Paris Law	Calculate steady state crack growth rates	
Region II and III Crack Growth	Differentiate steady state and unstable crack growth regimes Determine Kc as controlling factor for failure	
Example of Paris Law, Scatter, and Other Models	Determine number of cycles for steady state crack growth Recall typical scatter ranges for crack growth rates Recall hyperbolic sine model for capturing complete fatigue crack growth rate regimes	
Mean Stress Effects	Distinguish the mean stress effect on crack growth rates Evaluate common models for mean stress correction to the Paris Eq.	
Crack Closure Effects	Interpret the plasticity rationale for crack closure Evaluate the effective stress intensity range related to crack closure	
Sequencing Effects	Examine overloads related to plastic zone size, closure effects, and crack growth rates	
Characteristics of early stages of crack growth	Distinguish characteristics of Mode I and II cracking in stages I and II of fatigue crack growth Evaluate key micromechanical models for FCG: coarse slip model and blunting/sharpening model	HW 8

Microstructure features relation to crack growth rates	Examine grain size, barriers (grain boundaries), grain orientation, etc influence on crack growth rates	
Small crack growth	Examine crack size and variability of small fatigue crack growth rates	
Intergranular vs. Transgranular	Distinguish key characteristics of intergranular versus transgranular crack growth	
Kitagawa Diagram	Construct a Kitagawa diagram From a Kitagawa diagram, calculate finite/infinite life	
Breakdown of LEFM	Establish limitations for application of linear elastic fracture mechanics Classify small cracks	
Proof Tests	Identify the need and application of proof tests of components and structures	HW 9
8 - Sequencing Effects		
Definition of Variable Amplitude Loading	Define variable amplitude loading (VAL) and spectrum loading Infer the need for a methodology to translate constant amplitude fatigue information to VAL	
Linear Damage Quantification	Analyze the Palmgren-Miner (P-M) Linear Damage Equation	
Example of Palmgren-Miner Linear Damage Rule	Apply the P-M Eq to determine life for block loading spectrum	
Limitations of Linear Damage Rule and Sequencing Effects	Infer from physical rationale the importance of sequencing of loading Apply sequencing effects to accumulated damage laws	
Cycle Counting Approaches	Define cycle counting methods and attributes (peak, valley, range, reversals) to loading spectrum Recall the historical origins of cycle counting	
Rainflow Cycle Counting	Define protocol for rainflow counting method	
Example of Rainflow Cycle Counting to Segment Loading Cycles	Apply rainflow method to segment variable amplitude loading	
Example of Rainflow Cycle Counting applied to Lifing	Apply rainflow method to determine life	
Other Simple Cycle Counting Methods	Recall additional methods (Peak, Mean Crossing, Level Crossing, Range, Range-Mean, Fatigue-Meter, etc.) for cycle counting	
Variable Amplitude Loading applied to Strain-Life Approaches	Recall application of variable amplitude loading (rain flow) methods on strain-life approaches	
Variable Amplitude Loading applied to Fatigue Crack Growth Approaches	Build variable amplitude approaches based on plastic zone size Recall common (Elber, Equivalent K, Willenborg) models for VAL of fatigue crack growth	HW 10
9 - Sequencing Effects		
Definition of Corrosion Damage	Identify key attributes of corrosion damage	
Motivation for Studying Corrosion and General Effects of Corrosion	Identify fatigue debit and associated economic cost of corrosion damage Classify uniform versus local damage of corrosion	
Mechanisms of Corrosion Attack	Recall classifications of corrosion	
Uniform Corrosion, Pitting Corrosion, and Pit-to-Crack Transition	Recall key characteristics of uniform/pitting corrosion Distinguish mechanisms for transition of a corrosion pit to the formation/growth of a crack	
Galvanic corrosion	Establish chemo-mechanical mechanism of galvanic corrosion	
Selective Leaching, stress corrosion cracking, and intergranular corrosion	Recall key attributes of selective Leaching, stress corrosion cracking (SCC), and intergranular corrosion Identify need of sustained tensile stress, environment, and suceptible material to result in SCC	

Prevention of Corrosion and Life Extension	Identify four conditions that must be met for corrosion Identify measures for prevention (elimination of electrolyte or electrical contact) Establish approaches to perform life extensions for aircraft structures	
Hydrogen embrittlement	Summarize key mechanisms for hydrogen embrittlement and associated effect on fatigue	
Creep-Fatigue Interactions and Frequency Effects	Apply strain partitioning and frequency modification approaches to creep-fatigue interactions	
Dwell Effects	Identify two dwell phemenoma and physical rationale in aerospace materials: crack growth at high temperatures in Ni-based superalloys and load shedding during cold dwell of Ti alloys	
Thermo-Mechanical Effects for Constraints	Apply the effects of constraints on thermo-mechanical fatigue	
Load phases and damage mechanisms in thermo-mechanical fatigue	Evaluate isothermal, in-phase, out-of-phase loading conditions Distinguish fatigue, creep, oxidation damage mechanisms in thermo-mechanical fatigue	HW 11

5 Evaluation

Assignments	Due	Percentage
Homework (11x)	**Throughout the semester**	**25%**
Midterm	**10/21, 8-10 pm, MSEE B012**	**35%**
Final (cumulative)	**End of semester**	**40%**
		Total: 100%

In general, we will have a 90%|80%|70%|60% grade scale with +/- grades. Depending on how the class performs on its assignments and tests, the instructor reserves the right to curve the scale in the favor of the class, if necessary, based on his discretion. Grades will never be curved downward.

A+: 96.5-100%
A: 93.1-96.4%
A-: 89.5-93.0%
B+: 86.4-89.4%
B: 83.1-86.3%
B-: 79.5-83.0%
C+: 76.4-79.4%
C: 73.1-76.3%
C-: 69.5-73.0 %
D+: 66.4-69.4%
D: 63.1-66.3%
D-: 59.5-63.0%
F: 59.4% or below